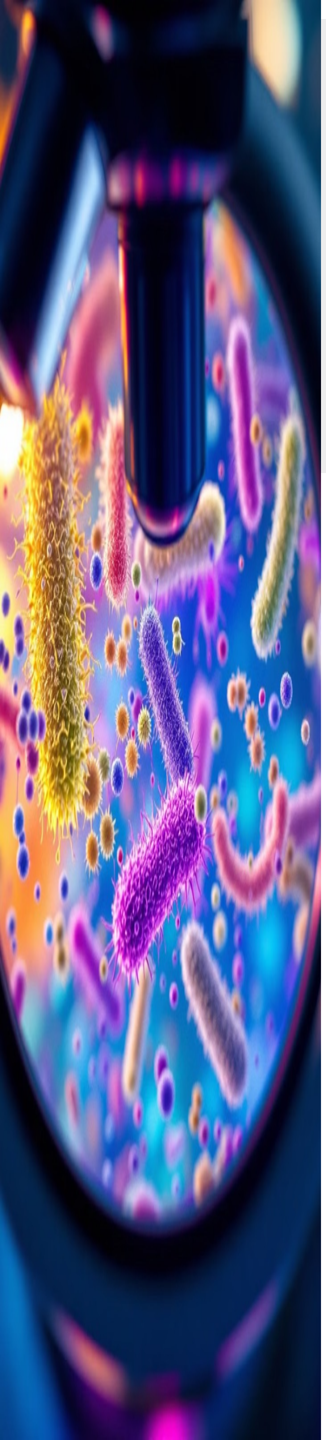
OCD-2O25-26

GOOD MORNING

INFECTION CONTROL IN DENTAL CARE SETTINGS

Dr. Vanaja Reddy
Associate professor

		LECTURE LE4ARNING OUTCOMES
L10- CLO		**By the end of this lecture, students should be able to:**
3.2	1	Describe the guidelines for infection control in dental unit
3.2	2	Illustrate steps in infection control in dental clinics



Microbiology Basics for Dental Personnel

Goal: Protect yourself, staff, and patients from illness.

Why Microbiology Matters

- Essential to understand infection risks.
- Encourages adherence to infection control protocols.

Types of Microorganisms

•**Pathogenic:** Cause disease.

•**Opportunistic Infections:** Affect individuals with weakened immune systems.

Major Groups of Microorganisms

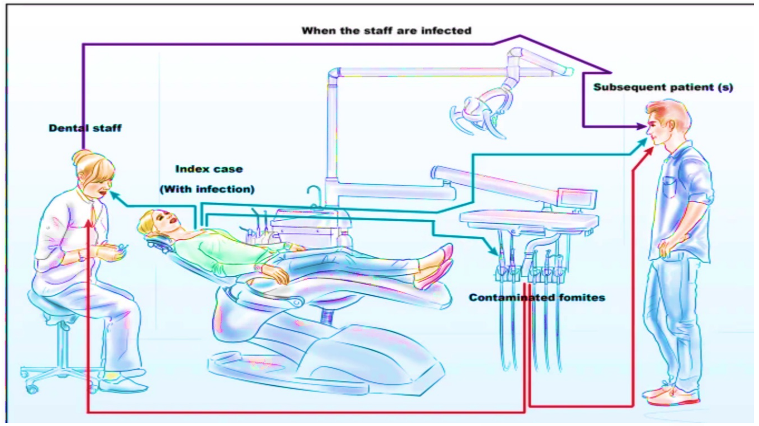
•**Bacteria:** strep throat , pneumonia, Tuberculosis etc.

•**Viruses:** flu, HIV, hepatitis etc.

•**Fungi:** Candidiasis, ringworm etc.

•**Protozoa:** Malaria ,toxoplasmosis etc.

How are Diseases Transmitted in the Dental Setting?



Patient → **Dental Health Care Personnel**

Dental Health Care Personnel → **Patient**

Patient → **Patient**

Modes of Transmission in Dental Settings

Direct Contact

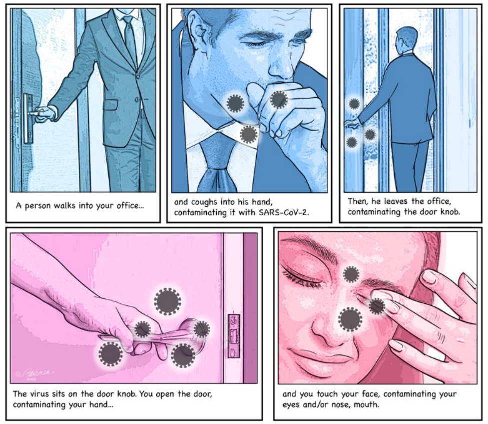
Touching an infected person or object.

Eg: Hand-to-hand contact, contact with contaminated instruments.

Indirect Contact

Touching a contaminated object.

Eg: Contact with contaminated surfaces, instruments, or materials.



Droplet Transmission

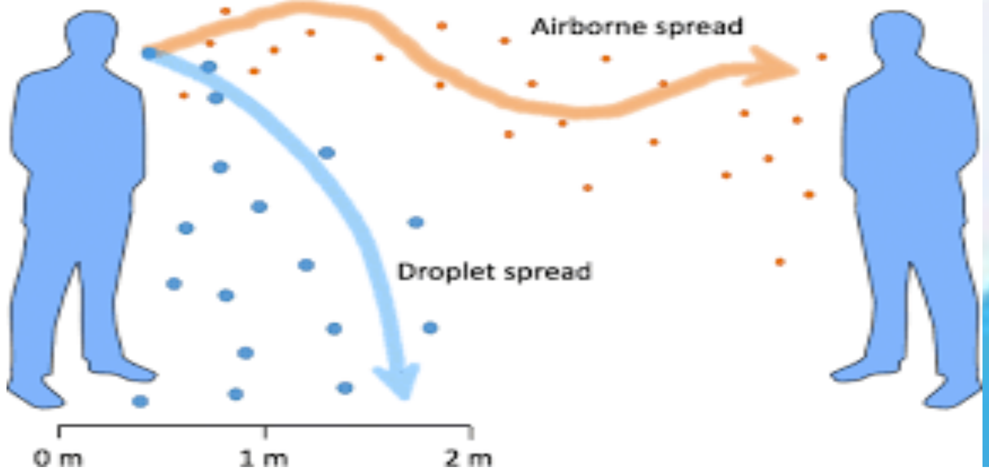
Inhaling droplets of saliva or respiratory secretions from an infected person.

Eg: Coughing, sneezing, talking, from sprays.

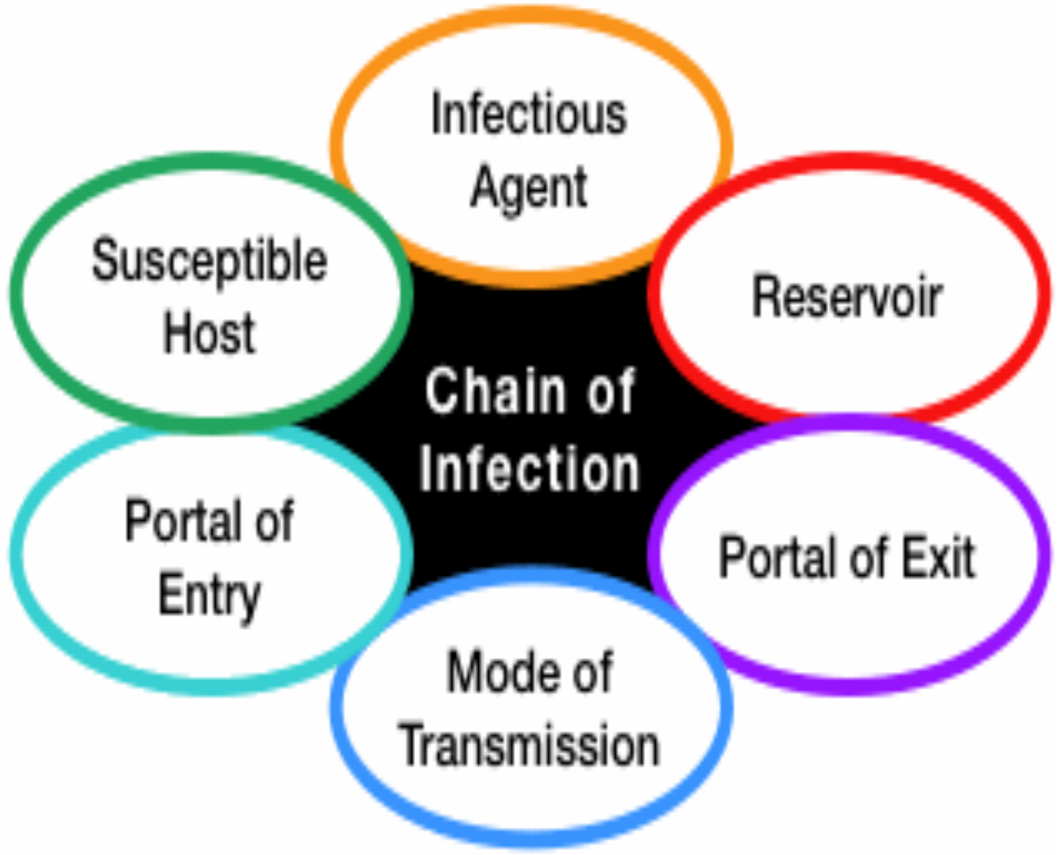
Airborne Transmission

Inhaling small particles of infectious agents that remain suspended in the air.

Eg: Aerosols from dental procedures, airborne particles from infected patients.



Chain of Infection



Example: Common cold spreading in a classroom

Infectious agent (Germ

Reservoir (Where the germ lives)
👉 The **infected student's nose and throat**

Transmission (How it spreads)
👉 The student **coughs or sneezes**, or touches desks/books with unwashed hands

Portal of entry (How it enters another person)
👉 Germs enter through **nose, mouth, or eyes** of another student.
(Breathing in droplets or touching face)

Susceptible host (Person who gets infected)
👉 A **healthy student** with low immunity or no protection

Goal of Infection Control

"Break the chain" of infection by consistently practicing Protocols which would Prevent the Infectious Agent from Moving to One Host to Another and Preventing Cross-contamination

http://www.cdc.gov/OralHealth/infectioncontrol/guidelines

Role of Key Organizations in Infection Control

- **CDC (Centers for Disease Control and Prevention)**

  Develops guidelines and recommendations for infection control in healthcare settings, including dentistry.

- **OSHA (Occupational Safety and Health Administration)**

  Enforces workplace safety standards, including those related to infectious diseases.

- **OSAP (Organization for Safety, Asepsis and Prevention)**

  Provides education, training, and resources for infection control in dentistry.

Chapter – 10. BURT, B. A. & EKLUND, S. A. 2005. Dentistry, Dental Practice and the Community, St. Louis: Elsevier Saunders 9 (pg: 139)

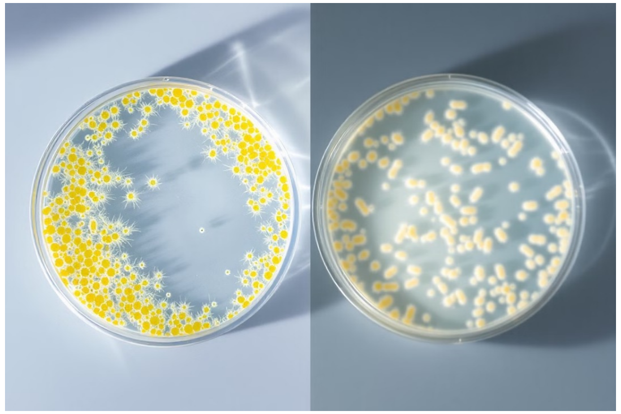
Bacteriostatic vs. Bacteriocidal

Bacteriostatic

Inhibits the growth of bacteria.
These agents do not kill bacteria, but they prevent them from multiplying.
Examples: Tetracycline, erythromycin.

Bacteriocidal

Kills bacteria.
These agents destroy bacteria, eliminating the infection.
Examples: Penicillin, amoxicillin.



Aerobic, Anaerobic, and Facultative Anaerobic Microorganisms

Aerobic: Require oxygen to survive and grow.
Examples: Staphylococcus aureus, Pseudomonas aeruginosa.

Anaerobic: Cannot survive in the presence of oxygen.
Examples: Clostridium tetani, Bacteroides fragilis.

Facultative Anaerobic: Can survive in the presence or absence of oxygen.
They can use oxygen for energy, but they can also survive without it.
Examples: Escherichia coli, Streptococcus pneumoniae.

➢ PRECAUTIONS

✓ Standard Precautions

A comprehensive approach to infection control that applies to all patients, regardless of their perceived infectious status.

It emphasizes the use of barriers, hand hygiene, and safe injection practices.

✓ Universal Precautions

A previous approach to infection control that treated all patients as potentially infected with HIV and Hepatitis B.

-Protect exposure from "Blood and other Body Fluids"

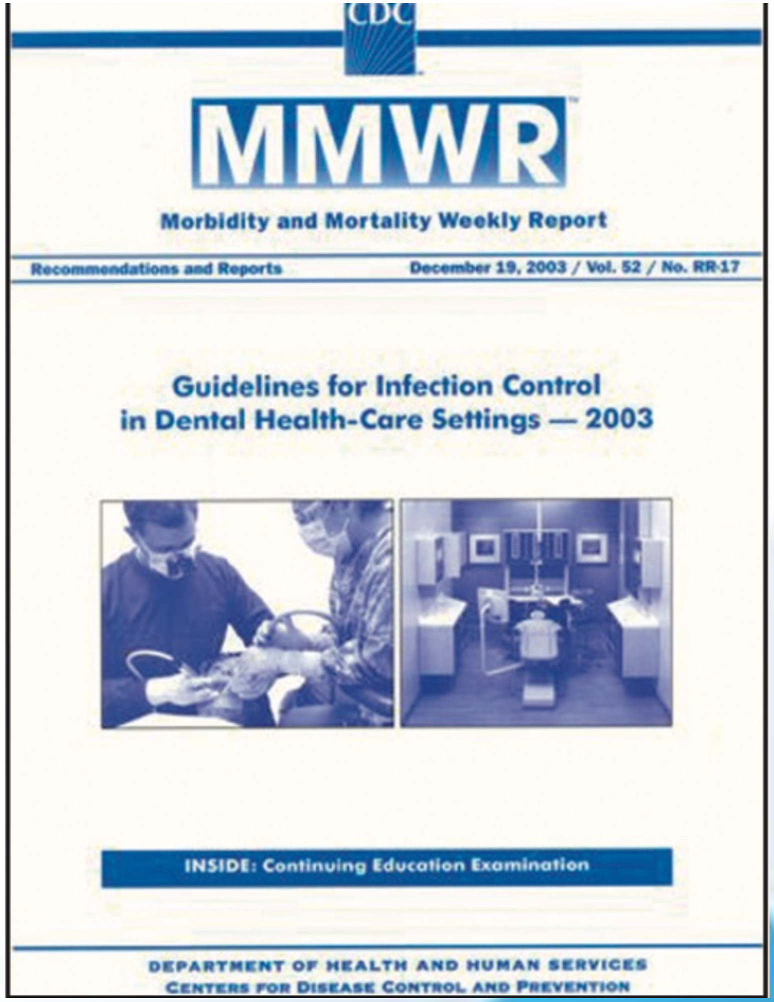
Infection Prevention Strategies

I. Engineering controls

II. Work practice controls

1. Performance of task
2. Hand hygiene
3. Personal protective equipment (PPE)
4. Cleaning surfaces
5. Waste management

III. Administrative controls



I. Engineering Controls

- Isolate or Remove the Hazard

Eg:

- Sharps container [Needle incinerator]
- Dental chairs with sterile water lines
- Medical devices with Injury Protection Features [e.g.self-sheathing needles]

II. Work practice control

1. Performance of task

Change the manner of performing task

Eg:

- Using Instruments instead of Fingers to Retract or Palpate Tissue
- One-handed needle recapping

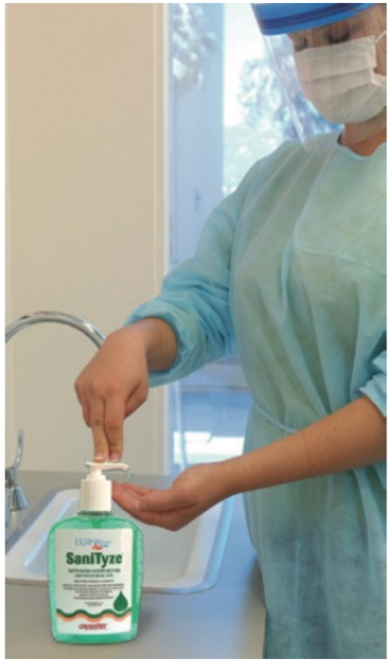
2. Hand Hygiene

a. Why Is Hand Hygiene Important?

- Hands are the Most Common mode of pathogen transmission
- Prevent health care-associated infections
- Reduce spread of Antimicrobial resistance

c.Hand Hygiene Methods

- Hand-washing
  - Washing hands with liquid soap and water
- Antiseptic Hand-wash
  - with water and Soap or other Detergents Containing an Antiseptic Agent (triclosan or chlorhexidine.)
- Alcohol-based Hand-rub
  - Rubbing hands with an Alcohol-containing preparation(waterless antiseptics)(60%–95% ethanol or isopropanol alcohol-containing)
- Surgical Antisepsis
  - with an antiseptic soap or an alcohol-based hand rub before operations by surgical personnel



Alcohol-based Preparations

Benefits

- Rapid and effective antimicrobial action
- Conc of 60-90%are more effective.

Limitations

- Cannot be used if hands are visibly soiled
- Inflammable: Store away from high temperatures

- **Special Hand Hygiene Considerations**
- Keep fingernails short
- Avoid artificial nails –can harbor pathogens.
- Avoid hand jewelry that may tear gloves
- Use hand lotions to prevent skin dryness

1

HAND WASHING

2

HAND -ALCOHOL RUB

3. Dispense the proper amount of the product into the palm of one hand.

4. Rub the palms of your hands together.

5. Rub the product between your fingers.

6. Rub the product over the back of your hands.
Purpose: It is important to thoroughly cover both of your hands.

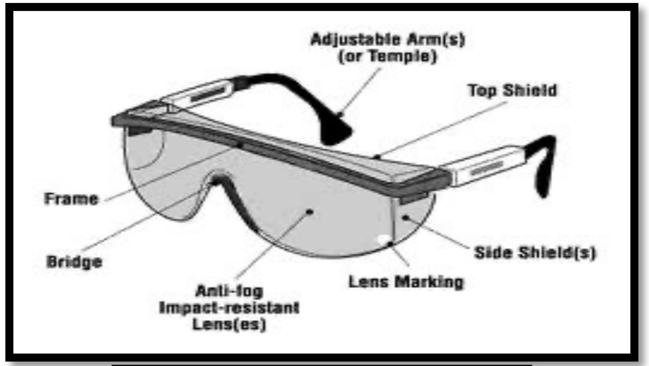
II. Work practice control

3.Personal Protective Equipment [PPE]

- A Major component of Standard Precautions
- Protects the skin and mucous membranes from exposure to infectious materials in spray or spatter
- Should be removed when leaving treatment areas

Masks, Protective Eyewear, Face Shields

Eyewear - protect the eyes against damage from aero-solized pathogens(Virus) ,from flying debris(as scrap amalgam and tooth fragments.)



-prevents injury from splattered solutions and caustic chemicals.

- Use of eyewear with both front and side protection (solid side shields)

-Protective eyewear that can be worn over prescription glasses is also available.

Face Shield to protect Mucous membranes of the Eyes, Nose, and Mouth

- **Guidelines for the Use of Protective Masks**

- Masks should be changed for every patient .

- Masks should be handled by touching only the side edges.

- Masks should not contact the mouth when worn because the moisture generated will decrease the mask filtration efficiency.

- Masks should not be worn below the nose or mouth

Personal Protective Equipment [PPE]

- Protective Clothing
- Wear gowns, lab coats, or uniforms that cover skin and personal clothing likely to become soiled with blood, saliva, or infectious material

Guidelines for the Use of Protective Clothing

- Because protective clothing can spread contamination, it is *not* worn out of the office for any reason, including travel to and from the office
- Protective clothing should be changed at least daily and more often if visibly soiled
- If a protective garment becomes visibly soiled or saturated with chemicals or body fluids, it should be changed immediately
- Protective clothing must *not* be worn in staff lounge areas or when workers are eating or consuming beverages

Remove all barriers before leaving the work area

7 Fluid-impervious gown. (Cc

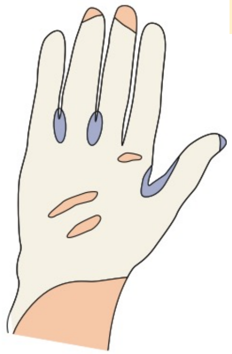
19-14 Appropriate clinical attire consists of long-sleevec gowns, gloves, and eyewear.

Personal Protective Equipment [PPE]

Gloves

- Minimize the risk of Health Care Personnel Acquiring Infections from Patients
- Prevent Microbial Flora from being transmitted from health care personnel to patients
- Are not a substitute for Handwashing!

- Recommendations for Gloving
- Wear gloves when contact with Blood, Saliva, and Mucous Membranes is possible
- Remove gloves after patient care
- Wear a new pair of gloves for each patient
- Remove gloves that are torn, cut or punctured
- Do not wash, disinfect or sterilize gloves for reuse



Personal Protective Equipment [PPE

WEARING STEPS

- ✔ Surgical mask
- ✔ Gloves

PROCEDURAL STEPS

1. Put on your protective clothing over your uniform, street clothes, or scrubs.
   Note: Protective clothing could be long-sleeved lab coats, clinic jackets, or gowns.

2. Put on your surgical mask, and adjust the fit.

4. Thoroughly wash and dry your hands.
   Note: If your hands are not visibly soiled, you may use an alcohol-based hand rub.
5. Hold one glove at the cuff, place your opposite hand inside the glove, and pull it onto your hand. Repeat with a new glove for your other hand.
   Important Note: Regarding the sequence of putting on PPE, the *most important* step is to put on the gloves last to avoid contaminating them before they are placed in the patient's mouth.

Modified from Policy to practice: OSAP's guide to the guidelines, Annapolis, MD, 2004, OSAP.

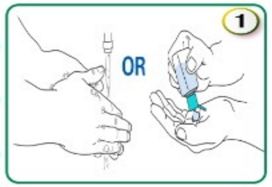
REMOVAL STEPS

وزارة الصحة
Ministry of Health

كيف ترتدي أدوات الحماية الشخصية
HOW TO PUT ON
PERSONAL PROTECTIVE EQUIPMENTS

قم بتنظيف اليدين جيداً

- يفضل استخدام معقم اليدين عندما لا تكون الأيدي متسخة بشكل واضح.
- نظف بين الأصابع ، ظهر اليدين ، أطراف الأصابع والإبهام.
- تنظيف الأيدي لمدة ٢٠-٣٠ ثانية.

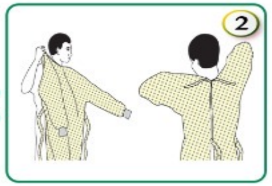


CLEAN YOUR HANDS

- Alcohol-based hand rub preferred method for cleaning hands where hands are not visibly soiled.
- Clean between fingers, backs of hands, fingertips & thumbs.
- Clean hands for 20-30 seconds.

قم بلبس سترة الوقاية

- اجعل فتحة سترة الوقاية من الخلف
- اربط السترة من الخلف حول الرقبة والخصر



PUT ON GOWN

- Keep openings to the back.
- Fasten in back of neck and waist

قم بلبس الكمامة

استخدام N95 من لديهم أمراض تنفسية أو عند اجراء (AEROSOL generating procedures)

PUT ON MASK

(Use N95 for AEROSOL generating procedures and in case of AIRBORNE precaution)

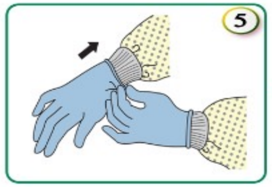
استخدام النظارات أو واقي الوجه

- قم بتغطية الوجه والعينين بالشكل الصحيح

PUT ON GOGGLES OR FACE SHIELD

- Place cover face and eyes and adjust to fit.

قم بلبس القفازات فوق سترة الوقاية



PUT ON GLOVES

- Extend to cover wrist of isolation gown.

استخدام ممارسات الأمنة لحماية نفسك والحد من انتشار التلوث :

. ابق يداك بعيدا عن الوجه
. الحد من ملامسة الأسطح
. تغيير القفازات عند تمزقها أو تلوثها بشدة
. نظف يديك

USE SAFE WORK PRACTICES TO PROTECT YOURSELF AND LIMIT THE SPREAD OF CONTAMINATION :

- Keep hands away from face
- Limit surfaces touched
- Change gloves when torn or heavily contaminated
- Perform hand hygiene

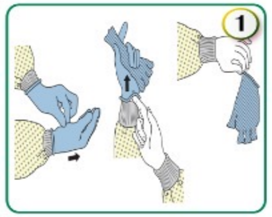
REFERENCE : وزارة الصحة Ministry of Health CDC

وزارة الصحة
Ministry of Health

كيف تخلع أدوات الحماية الشخصية
HOW TO REMOVE
PERSONAL PROTECTIVE EQUIPMENTS

قم بخلع القفازات

- في حال تلوث اليدين عند إزالة القفاز قم فوراً بغسل اليدين أو استعمال معقم اليد الكحولي
- التخلص من القفازات في حاوية النفايات (الصفراء).

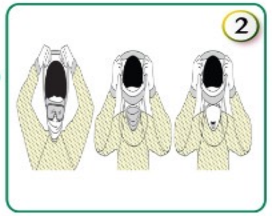


TAKE OFF GLOVES UNDER THE WRIST

- If your hands get contaminated during glove removal, immediately wash your hands or use an alcohol-based hand sanitizer
- Discard gloves in a yellow waste container

أخلع نظارات الوقاية أو درع الوجه

- إزالة نظارات الوقاية أو درع الوجه من الخلف وعدم لمس الجهة الامامية



TAKE OFF GOGGLES OR FACE SHIELD

- Remove goggles or face shield from the back and not touch the front

اخلع سترة الوقاية

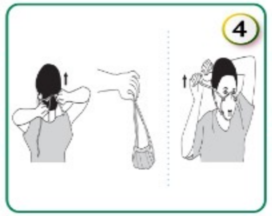
- توخى الحذر عند إزالة سترة الوقاية .
- اسحب سترة الوقاية بعيداً عن العنق والكتفين ، ولمس داخل الثوب فقط .

TAKE OFF GOWN

- Take care when remove gown
- Pull gown away from neck and shoulders, touching inside of gown only

اخلع الكمامة

- ارفع الجزء السفلي للكمامة بمرونة فوق رأسك أولاً. ثم ارفع الجزء الاعلى بمرونة.
- ابقها بعيداً عن الوجه .
- تخلص منها في حاوية النفايات الصفراء .



TAKE OFF THE MASK

- Lift the bottom elastic over your head first. Then, lift off the top elastic.
- Lift away from face
- Discard in a yellow waste container

قم بغسل اليدين جيداً

- يفضل استخدام معقم اليدين عندما لا تكون الأيدي متسخة بشكل واضح.
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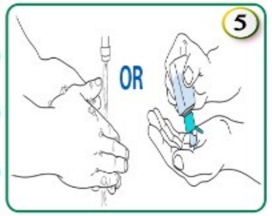

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- Clean hands for 20-30 seconds.

REFERENCE : وزارة الصحة Ministry of Health CDC

II. Work practice control

4. Cleaning surfaces

Clinical Contact Surfaces

- Light handle
- Countertop
- Bracket tray
- Dental chair
- Door handle

Cleaning Clinical Contact Surfaces

- Surface barriers can be used and changed between patients
- Clean then disinfect using an "Environment Protection Agency"[EPA] registered hospital disinfectant.

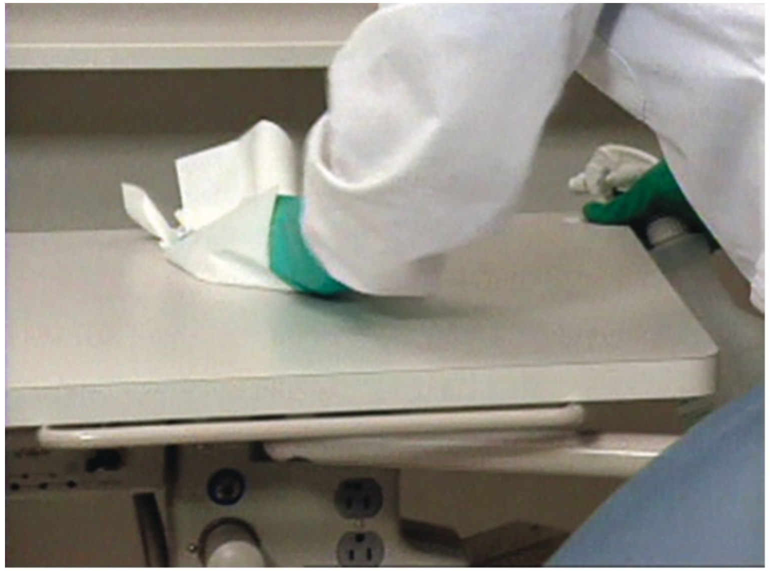

FIGURE 20-2 Smooth surfaces are easily sprayed and wiped.

House-keeping Surfaces

- Walls
- Sinks
- floors

- Routinely clean with soap and water or an EPA-registered detergent/hospital disinfectant routinely
- Clean mops and cloths and allow to dry thoroughly before re-using
- Prepare fresh cleaning and disinfecting solutions daily and per manufacturer recommendations

FIGURE 20-5 Rolls of plastic tubing that can be cut to the desired length. (Courtesy Certol, Commerce City, CO.)

IGURE 20-4 Surfaces touched during patient care should be ›vered with protective barriers. If not protected, they should be eaned and disinfected at the end of the procedure.

FIGURE 20-6 Tube socks provide barrier protection for difficult-to-clean areas. (Courtesy Certol, Commerce City, CO.)

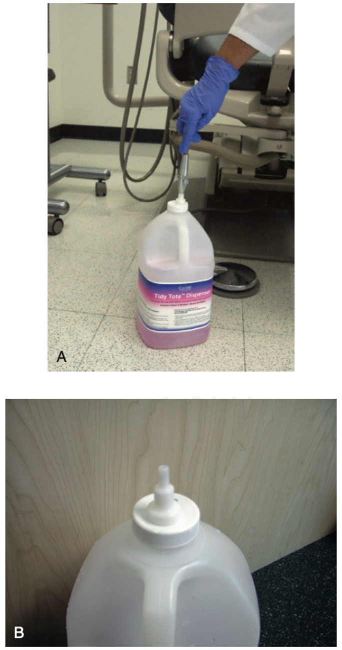

FIGURE 20-12 **A,** ProE-Vac liquid evacuation system cleaner. **B,** Convenient closed container for transporting solution between units. (Courtesy Certol, Commerce City, CO.)

II. Work practice control

5. Waste management

- **Medical waste** is all **waste** materials generated at health care facilities: regulated and non-regulated.

Regulated Medical Waste: Poses a potential risk of infection during handling and disposal

- solid waste soaked or saturated with blood or saliva (e.g., gauze saturated with blood after surgery),
- extracted teeth,
- surgically removed hard and soft tissues, and
- contaminated sharp items (e.g., needles, scalpel blades, and wires)

- Medical wastes are "treated" in accordance with state and local EPA regulations

- Processes for regulated waste include autoclaving and incineration

Color-coded waste containers

Color Coding Recommended for Waste Bags or Containers

Type of Waste	Color Case or Container
Highly infectious waste (such as culture plates)	Orange clearly marked «HIGHLY INFECTIOUS» (Arabic and English) with biohazard logo. Leak-proof and strong plastic bag, or container for autoclaving
Infectious waste	Orange clearly marked «Hazardous Waste» (Arabic and English) with biohazard logo Leak-proof plastic bag, or container
Human and animal body parts and organs	Red clearly marked «Hazardous Waste» (Arabic and English) with biohazard logo Leak proof plastic bag or container
Sharps	Yellow clearly marked « Hazard - Sharp Items « (Arabic and English) with biohazard logo Puncture-proof containers
Non-risk healthcare waste	Black Plastic bag or container

https://dent.ksu.edu.sa/sites/dent.ksu.edu.sa/files/attach/infection_control_guidelines.pdf

III.Administrative Controls

- Policies, Procedures, and Enforcement measures

- Placement in the hierarchy varies by the problem being addressed
    - Placed before engineering controls for airborne precautions [e.g. TB]

- **<u>Infection control in Dental Radiology</u>**
- Prevent cross-contamination with saliva or blood by wearing gloves when taking radiographs and handling contaminated film packets.
- Use personal protective equipment (PPE) if there is a risk of saliva spattering.
- Heat sterilize or disinfect semi-critical items like film holders with high-level disinfection methods.
- Dry the film with gauze to remove blood or saliva before removing gloves.
- If film pouches are used, carefully remove the film /PSP to avoid contamination

- Avoid contamination of developing equipment.
- Non critical (Xray tubehead, control panel) should be protected by surface barriers or shoud be disinfected after every use.

Checklist for Infection Control in Dental Radiography

Before Exposure

Treatment Area (Covered or Disinfected)

- X-ray machine
- Dental chair
- Work area
- Lead apron
- Computer keyboard/mouse

Equipment and Supplies (Prepared Before Seating Patient)

- Image receptor (film, sensor, or phosphor storage plate [PSP])
- Positioning devices
- Cotton rolls
- Paper towel
- Disposable container

Patient Preparation (Performed Before Putting on Gloves)

1. Adjust chair.
2. Adjust headrest.
3. Place lead apron.
4. Remove personal objects.

Operator Preparation (Completed Before Exposure)

1. Wash hands.
2. Put on gloves.
3. Assemble positioning devices.

During Exposure

Film/PSP Handling

1. Dry exposed film or PSP with paper towel.
2. Place dried film in disposable container or transfer box.

Positioning Devices

1. Transfer positioner from work area to mouth.
2. Transfer positioner from mouth back to work area.

   Note: Never place a positioner on uncovered countertop.

After Exposure

Before Glove Removal

1. Dispose of all contaminated items.
2. Place positioning devices in designated area for contaminated equipment.

After Glove Removal

1. Wash hands.
2. Remove lead apron.

E 40-5 X-ray exposure control with barriers in place.

URE 40-6 Radiography operatory with barriers in plac

FIGURE 40-10 Protective packet barrier over phosphor stor plate (PSP).

FIGURE 40-11 FDA-cleared sensor and cable barriers. (Court Dentsply Rinn, Elgin, IL.)

REFERENCES:

- Guidelines for Infection Control in Dental Health-Care Settings,2003 by CDC.
- Student Infection Control Manual of Dentistry Program of Ibn Sina National College
- Modern dental assisting 11th edition.

Thank you